

1. Context

- (a) Overview of Quantum Computing
- (b) Physics of Periodic Structures
 - i. Density functional theory
 - ii. Tight-binding model
 - iii. Variational effective mass approximation
- (c) Historical narrative of EMA
 - i. Kohn-Luttinger
 - ii. Faulkner
 - iii. Shino-Nara
 - iv. Gamble

2. Analytical Model

- (a) Derivation of EMA energy functional

Shallow semiconductor heterostructures in the bulk

The interruption of the crystalline structure of the semiconductor, as with impurities or heterostructures, can serve to capture excess electrons within the semiconductor. We wish to examine the physics of these captured electrons. The Hamiltonian of such a system may be written as

$$\left[\frac{\hat{\mathbf{p}}^2}{2m_e} + \hat{V}^0 + \hat{U} \right] |\Psi\rangle = E |\Psi\rangle, \quad (1)$$

where $\hat{V}^0$ is the periodic potential of the crystalline lattice, $\hat{U}$ is the potential introduced by a defect in the lattice through an impurity or other means, and E has energy zero at the lowest conduction band minima.

We understand that, in the absence of defect, the wavefunction resulting from the periodic crystal potential will be a Bloch function. The acceptable energies for free electrons occur in discrete bands that may be separated, resulting in disallowed energies. Each Bloch function is denoted by a wavevector and an

index indicating membership in one of these bands. The wavefunction of each Bloch state follows the form

$$\langle \mathbf{r} | n\mathbf{k} \rangle = \varphi_{n\mathbf{k}}(\mathbf{r}) = \exp(i\mathbf{k} \cdot \mathbf{r}) u_{n\mathbf{k}}(\mathbf{r}), \quad (2)$$

where $u_{n\mathbf{k}}$ may be a complicated function, but one which always has the periodicity of the lattice. The Bloch state is the eigenstate of a nearly free electron travelling through a silicon lattice. As such, the Schrödinger equation of the pure system may be written as

$$\left[\frac{\hat{\mathbf{p}}^2}{2m_e} + \hat{V}^0 \right] |n\mathbf{k}\rangle = E_{n\mathbf{k}} |n\mathbf{k}\rangle \quad (3)$$

V^0 is the periodic potential of silicon and $E_{n\mathbf{k}}$ is the energy of a nearly free electron travelling through silicon in the n th conduction band given the wavevector k .

We assume that Bloch functions taken from the first Brillouin zone and from all bands form a complete basis to be used to construct the state of an excess electron bound to an impurity or heterostructure. As such, we represent the complete wavefunction as

$$|\Psi\rangle = \sum_n \int_{BZ} d\mathbf{k} A_n(\mathbf{k}) |n\mathbf{k}\rangle \quad (4)$$

We write the Schrödinger equation in the Bloch basis, letting U be the impurity potential,

$$(E_{n\mathbf{k}} - E) A_n(\mathbf{k}) + \sum_{n'} \int_{BZ} d\mathbf{k}' A_{n'}(\mathbf{k}') \langle n\mathbf{k} | \hat{U} | n'\mathbf{k}' \rangle = 0 \quad (5)$$

We can clearly see the role of the impurity potential in mixing together different bands, as well as causing the state to spread through each band. However, this equation proves difficult to solve directly, since it relies on the interplay between the Fourier components of the impurity potential and the periodic function $u_{n\mathbf{k}}$. To resolve these difficulties we employ a variational method. In the variational method one calculates $E[\Psi]$ from

$$\langle \Psi | \frac{\hat{\mathbf{p}}^2}{2m_e} + \hat{V}^0 + \hat{U} - E[\Psi] | \Psi \rangle = 0, \quad (6)$$

where the state Ψ is judiciously chosen, with variational parameters, to solve the problem approximately. Minimization of the energy with respect to the parameters provides the best possible approximate solution to the problem using

the state Ψ . Reformulating the Schrödinger equation in light of the variational method yields

$$\sum_n \int_{BZ} d\mathbf{k} |A_n(\mathbf{k})|^2 (E_{n\mathbf{k}} - E) + \sum_{n,n'} \iint_{BZ} d\mathbf{k} d\mathbf{k}' A_n(\mathbf{k})^* A_{n'}(\mathbf{k}') \langle n\mathbf{k} | \hat{U} | n'\mathbf{k}' \rangle = 0. \quad (7)$$

Shallow donor electron states

At this point, we explicitly restrict our concern to “shallow” potentials. For our purposes, the “shallowness” of a potential depends on its effect in mixing and occupying the free electron states. Namely, the bound state of a shallow potential only significantly occupies the lowest conduction band and occupation within that band is centered steeply around any minima. As such, the bound state possesses non-periodic structure that emerges at a scale much longer than the characteristic wavelength of the Bloch function at the conduction band minima. The shallowness assumptions leads to a series of conclusions:

- i. The potential does not cause significant band occupation outside the lowest conduction band, and so terms $n \neq 0$ may be ignored.
- ii. The band occupation will mostly occur in “valleys” – the k -space energy minima of the lowest conduction band. As such, we write the occupation as $A_0(\mathbf{k}) = \sum_q \alpha_q A^{(q)}(\mathbf{k})$ where the coefficients α_q are determined from group theoretic symmetry concerns and $A^{(q)}(\mathbf{k})$ are sharply peaked functions centered at the q th minimum of the conduction band.

Because we expect the $A^{(q)}(\mathbf{k})$ to be sharply peaked at a particular valley in the conduction band, we can expand $E_n(\mathbf{k})$ in Eq. (7) up to the second order in $\mathbf{k}$ at each valley. Since the valleys are minima, the energy will only contain second order terms, making it analogous to the kinetic energy. This analogy is the source of the effective mass that gives the approximation its name. In general, the minima is anisotropic in $\mathbf{k}$, leading to multiple effective masses.

We wish to see the effect of these assumptions upon Eq. (7). The diagonal part

is straight forward, and so we can write

$$\langle \Psi | \frac{\hat{\mathbf{p}}^2}{2m_e} + \hat{V}^0 - E | \Psi \rangle = \sum_q |\alpha^{(q)}|^2 \int_{BZ} d\mathbf{k} |A^{(q)}(\mathbf{k})|^2 (E^{(q)}(\mathbf{k}) - E) \quad (8)$$

where $E^{(q)}(\mathbf{k})$ is the second order expansion of $E_0(\mathbf{k})$ around the q th valley. Turning our concern to the impurity term in Eq. (7) yields

$$\langle \Psi | \hat{U} | \Psi \rangle = \sum_{q,q'} \alpha_q^* \alpha_{q'} \iint_{BZ} d\mathbf{k} d\mathbf{k}' A^{(q)}(\mathbf{k})^* A^{(q')}(\mathbf{k}') \langle 0\mathbf{k} | \hat{U} | 0\mathbf{k}' \rangle, \quad (9)$$

The matrix element is computed as as

$$\langle 0\mathbf{k} | \hat{U} | 0\mathbf{k}' \rangle = \int d\mathbf{r} U(\mathbf{r}) e^{i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{r}} u_{0\mathbf{k}}^*(\mathbf{r}) u_{0\mathbf{k}'}(\mathbf{r}), \quad (10)$$

so I can write the impurity term as

$$\langle \Psi | \hat{U} | \Psi \rangle = \sum_{q,q'} \alpha_q^* \alpha_{q'} \int d\mathbf{r} U(\mathbf{r}) \iint_{BZ} d\mathbf{k} d\mathbf{k}' A^{(q)}(\mathbf{k})^* e^{-i\mathbf{k} \cdot \mathbf{r}} u_{0\mathbf{k}}^*(\mathbf{r}) A^{(q')}(\mathbf{k}') e^{i\mathbf{k}' \cdot \mathbf{r}} u_{0\mathbf{k}'}(\mathbf{r}) \quad (11)$$

If we wish to examine the long scale physics, it's convenient to introduce a shifted inverse truncated Fourier transform of the occupation function,

$$F^{(q)}(\mathbf{r}) = \frac{1}{(2\pi)^{3/2}} \int_{BZ} d\mathbf{k} A^{(q)}(\mathbf{k}) e^{i(\mathbf{k} - \mathbf{k}_q) \cdot \mathbf{r}} \quad (12)$$

which is often called the “envelope function” since it predicts the long-scale behavior of the wavefunction. With this definition of the envelope function and because $A^{(q)}(\mathbf{k})$ is so steeply peaked about the valley minima, we approximate write the $\mathbf{k}$ -space integrals in the impurity term as

$$F^{(q)}(\mathbf{r}) e^{i\mathbf{k}_q \cdot \mathbf{r}} u_{0\mathbf{k}_q}(\mathbf{r}) \approx \frac{1}{(2\pi)^{3/2}} \int_{BZ} d\mathbf{k} A^{(q)}(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{r}} u_{0\mathbf{k}}(\mathbf{r}) \quad (13)$$

With this approximation, we find the impurity term is

$$\langle \Psi | \hat{U} | \Psi \rangle = (2\pi)^3 \sum_{q,q'} \alpha_q^* \alpha_{q'} \int d\mathbf{r} U(\mathbf{r}) F^{(q)}(\mathbf{r})^* e^{-i\mathbf{k}_q \cdot \mathbf{r}} u_{0\mathbf{k}_q}^*(\mathbf{r}) F^{(q')}(\mathbf{r}) e^{i\mathbf{k}_{q'} \cdot \mathbf{r}} u_{0\mathbf{k}_{q'}}(\mathbf{r}) \quad (14)$$

We can also reexpress the diagonal part of the variational Hamiltonian using the envelope function. I introduce an effective kinetic energy operator for each valley $\hat{T}^{(q)}$ such that

$$\hat{T}^{(q)} F^{(q)}(\mathbf{r}) = \frac{1}{(2\pi)^3} \int_{BZ} d\mathbf{k} E^{(q)}(\mathbf{k}) A^{(q)}(\mathbf{k}) e^{i(\mathbf{k} - \mathbf{k}_q) \cdot \mathbf{r}}. \quad (15)$$

As well, it can be shown that

$$\int_{BZ} d\mathbf{k} |A^{(q)}(\mathbf{k})|^2 = \int d\mathbf{r} |F^{(q)}(\mathbf{r})|^2 \quad (16)$$

This allows us to restate equation (8) as

$$\langle \Psi | \frac{\hat{\mathbf{p}}^2}{2m_e} + \hat{V}^0 - E | \Psi \rangle = \sum_q |\alpha^{(q)}|^2 \int d\mathbf{r} F^{(q)}(\mathbf{r})^* \left(\hat{T}^{(q)} - E \right) F^{(q)}(\mathbf{r}) \quad (17)$$

With the state given in terms of the envelope function we can combine Eqs. (14) and (17) as the following variational equation, which may be solved explicitly for the functional expectation energy E ,

$$E = \sum_{q,q'} \alpha_q^* \alpha_{q'} \left(\delta_{qq'} T^{(q)} + U^{(qq')} \right), \quad (18)$$

where

$$T^{(q)} = \int d\mathbf{r} F^{(q)}(\mathbf{r})^* \hat{T}^{(q)} F^{(q)}(\mathbf{r}), \quad (19)$$

$$U^{(qq')} = \int d\mathbf{r} (2\pi)^3 u_{0\mathbf{k}_q}^*(\mathbf{r}) u_{0\mathbf{k}_{q'}}(\mathbf{r}) e^{i(\mathbf{k}_{q'} - \mathbf{k}_q) \cdot \mathbf{r}} F^{(q)}(\mathbf{r})^* F^{(q')}(\mathbf{r}) U(\mathbf{r}), \quad (20)$$

$$\sum_q |\alpha_q|^2 \int d\mathbf{r} |F^{(q)}(\mathbf{r})|^2 = 1. \quad (21)$$

The concern turns now to an adequate choice of functions $F^{(q)}$ so that the variational method may yield adequate results. The choice of $F^{(q)}$ will depend strongly on the band structure of the hosting material. In the absense of information about the host material, we can still represent the envelope function by expanding it in a known basis. As such, we write

$$F^{(q)}(\mathbf{r}) = \sum_i c_i^{(q)} \Phi_i^{(q)}(\mathbf{r}; \Omega_i^{(q)}) \quad (22)$$

We equip our basis with a set of variational parameters Ω to better explore the state space and to minimize the spectrum. If we insert (22) into (18) we are able to write our variational equation as

$$E(\Omega_1, \Omega_2) \sum_{iq, (iq)'} \beta_{iq}^* \beta_{(iq)'} S_{ii'}^{(qq')} = \sum_{iq, (iq)'} \beta_{iq}^* \beta_{(iq)'} \left(T_{ii'}^{(qq')} + U_{ii'}^{(qq')} \right) \quad (23)$$

where S , T , and U are understood as matrices with elements given as

$$S_{ii'}^{(qq')} = \delta_{qq'} \int d\mathbf{r} \Phi_i^{(q)*} \Phi_{i'}^{(q)} \quad (24)$$

$$T_{ii'}^{(qq')} = \delta_{qq'} \int d\mathbf{r} \Phi_i^{(q)*} \hat{T}^q \Phi_{i'}^{(q)} \quad (25)$$

$$U_{ii'}^{(qq')} = \int d\mathbf{r} (2\pi)^3 u_{0\mathbf{k}_q}^* u_{0\mathbf{k}_{q'}} e^{i(\mathbf{k}_{q'} - \mathbf{k}_q) \cdot \mathbf{r}} \Phi_i^{(q)*} \Phi_{i'}^{(q')} U(\mathbf{r}) \quad (26)$$

$$\beta_{iq} = c_i^{(q)} \alpha_q \quad (27)$$

We solve the variational equation above by solving the associated generalized eigenvalue problem,

$$E(\Omega_1, \Omega_2) \mathbf{S} \boldsymbol{\beta} = (\mathbf{T} + \mathbf{U}) \boldsymbol{\beta}, \quad (28)$$

where E and $\boldsymbol{\beta} = (c_i^{(q)} \alpha_q)$ is the generalized eigenvalue and -vector, respectively. The eigenvector may be normalized by satisfying

$$\boldsymbol{\beta}^T \mathbf{S} \boldsymbol{\beta} = 1. \quad (29)$$

From the generalized eigenvalue problem, we acquire information both about the energy and the state associated with a state of variational parameters, and we can write such a state, then, in terms of our basis and our variational parameters as

$$\langle \mathbf{r} | \Psi(\Omega) \rangle \approx \sum_{iq} \beta_{iq} \Phi_i^{(q)} \left(\mathbf{r}; \Omega_i^{(q)} \right) e^{i\mathbf{k}_q \cdot \mathbf{r}} (2\pi)^{3/2} u_{\mathbf{k}_q}(\mathbf{r}). \quad (30)$$

- (b) Variational basis, defined and manipulated
 - i. SubBasis is complete, extensible, variational.
 - ii. SuperBasis covers spectrum in state space.
- (c) Symmetry concerns
 - i. Symmetrizing basis states
 - ii. Symmetrizing potential
 - iii. Symmetry and computation of matrix element integrals
- (d) Incorporating band-structure and Bloch functions.
 - i. Symmetry and the Bloch functions
- (e) Considering the effective mean field potential.

- i. Statically screened Coulomb potential
- ii. Dynamically screened Coulomb potential
- iii. Center-cell corrections
 - A. Symmetry and bond distortion
 - B. Potential cutoff and exchange-correlation

3. Computational Description

4. Results

- (a) Statically screened Coulomb potential
- (b) Dynamically screened Coulomb potential
- (c) Center-cell corrections
 - i. Symmetry and bond distortion
 - ii. Potential cutoff and exchange-correlation

5. Appendix

- (a) Calculation of matrix elements
- (b) T_d point-group symmetry and selection rules
- (c) Hypergeometric functions